

Sec 5.5 Nonhomog Eqs (method of undeter. coeffs part 2)

"Problem Case" $y'' - y' - 2y = e^{2x}$

$$y_c = c_1 e^{-x} + c_2 \boxed{e^{2x}}$$

• since $f(x) = e^{2x} \Rightarrow$ trial $y_p = Ae^{2x}$

$$y_p' = 2Ae^{2x}$$

$$y_p'' = 4Ae^{2x}$$

plug in to check $y_p'' - y_p' - 2y_p = e^{2x} ?$

$$4Ae^{2x} - 2Ae^{2x} - 2Ae^{2x} \stackrel{?}{=} e^{2x}$$

$$(4A - 4A)e^{2x} = e^{2x}$$

$$0 = e^{2x} \dots \therefore \text{trouble.}$$

What went wrong?

• $y_p = Ae^{2x}$ is multiple of one of the "basis solutions" in y_c , which by defin must satisfy $y'' - y' - 2y = 0$.

so trying y_p , we got $(Ae^{2x})'' - (Ae^{2x})' - 2(Ae^{2x})$

But this is $A \left(\underbrace{(e^{2x})'' - (e^{2x})' - 2(e^{2x})}_{\text{this} = 0} \right)$

this = 0

(b/c e^{2x} satisfies $y'' - y' - 2y = 0$)

in summary, problem is that y_p has a piece that is not linearly indep. from the "basis functions" " y_1, y_2 " that make up y_c

$$y_c = c_1 \underbrace{e^{-x}}_{y_1} + c_2 \underbrace{e^{2x}}_{y_2}$$

problem trialing $y_p = Ae^{2x} = Ay_2$

so to fix this dependence, we change our trial function $y_p = Ae^{2x} \longrightarrow \underline{y_p = A \cancel{x} e^{2x}}$
new trial

note that now y_p is linearly indep from $y_1 = e^{-x}, y_2 = e^{2x}$.

so now $y_p = \underline{Ax e^{2x}}$

$$y_p' = Ae^{2x} + 2Ax e^{2x} = \underline{A(\cancel{1} + 2x)e^{2x}}$$

$$y_p'' = 2Ae^{2x} + A(1+2x) \cdot 2e^{2x} \\ = 2A(2+2x)e^{2x} = \underline{4A(1+\cancel{1}x)e^{2x}}$$

plug in to $y'' - y' - 2y = e^{\overset{2}{\cancel{3}}x}$

$$4A(1+x)e^{2x} - A(1+2x)e^{2x} - 2(Ax)e^{2x} = e^{2x}$$

$$e^{2x}(4A - A) + x e^{2x}(\cancel{4A} - \cancel{2A} - \cancel{2A}) = e^{2x} \\ 3A e^{2x} = e^{2x} \Rightarrow \boxed{A = 1/3}$$

$$\text{So } y_p = A x e^{2x} = \frac{1}{3} x e^{2x}$$

Step 3 combine y_c , y_p

General solution for $y'' - y' - 2y = e^{2x}$

is $y = y_c + y_p$

$$\boxed{y = c_1 e^{-x} + c_2 e^{2x} + \frac{1}{3} x e^{2x}}$$

recap: $y'' - y' - 2y = e^{2x}$

- $y_c = c_1 \boxed{e^{-x}} + c_2 \boxed{e^{2x}} = c_1 y_1 + c_2 y_2$
- $f(x) = e^{2x}$
- "original trial" $y_p = A \boxed{e^{2x}}$ had "duplication" with (part of) y_c , specifically, y_p was not lin. ind. from $y_2 = e^{2x}$. ("duplication")
- To "remove duplication" we changed y_p
 $y_p = A e^{2x} \longrightarrow y_p = A \underline{x} e^{2x}$
and then used this new y_p as trial.
- We solved for $A \dots$, found $y_p = \frac{1}{3} x e^{2x}$
- Finally we combined $y = y_c + y_p$
 $= c_1 e^{-x} + c_2 e^{2x} + \frac{1}{3} x e^{2x}$.

Hyperbolic trig functions



$$\cosh(x) \stackrel{\text{DEF}}{=} \frac{e^x + e^{-x}}{2}$$

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

$$\sinh(x) \stackrel{\text{DEF}}{=} \frac{e^x - e^{-x}}{2}$$

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

"vice versa" have $\cdot e^x = \cosh(x) + \sinh(x)$
 $\cdot e^{-x} = \cosh(x) - \sinh(x)$

$$\left[\begin{array}{l} \frac{d}{dx} \cosh(x) = \sinh(x) \\ \frac{d}{dx} \sinh(x) = \cosh(x) \end{array} \right]$$

$\rightarrow$ $\cosh, \sinh$ are two lin indep solutions of $y'' = y$

Ex: $y'' - y = \cosh(x)$

Step 1: y_c : char eq $r^2 - 1 = 0 \Rightarrow r = \pm 1$,

so gen. solution is $y_c = c_1 \boxed{e^{-x}} + c_2 \boxed{e^x}$

Step 2 (Option 1): Changing $\cosh, \sinh$ to exponentials

$$f(x) = \cosh x = \frac{1}{2} e^{-x} + \frac{1}{2} e^x$$

so think trial $y_p = A \boxed{e^{-x}} + B \boxed{e^x}$

need to check for potential duplication with y_c .

$\cdot$ Both e^{-x} and e^x have duplication with y_c

 so we change $e^{-x} \rightarrow xe^{-x}$, $e^x \rightarrow xe^x$ for y_p

So new trial $y_p = \underline{Axe^{-x}} + \underline{Bxe^x}$
no duplication with y_c now.

$$y_p' = Ae^{-x} - Axe^{-x} + Be^x + Bxe^x \\ = A(1-x)e^{-x} + B(1+x)e^x$$

$$y_p'' = -Ae^{-x} - A(1-x)e^{-x} + Be^x + B(1+x)e^x \\ (-2A + Ax)e^{-x} \\ A(-2+x)e^{-x} + B(2+x)e^x$$

Orig ODE was $y'' - y = \cosh(x)$

$$y'' - y = \frac{1}{2}e^{-x} + \frac{1}{2}e^x$$

plug in y_p :

$$A(-2+x)e^{-x} + B(2+x)e^x \\ - Axe^{-x} - Bxe^x$$

$$e^{-x}(-2A) + xe^{-x}(A-A) + e^x(2B) + xe^x(B-B)$$

$$-2Ae^{-x} + 2Be^x \stackrel{\text{(want)}}{=} \frac{1}{2}e^{-x} + \frac{1}{2}e^x$$

$$\Rightarrow \begin{cases} -2A = \frac{1}{2} \Rightarrow A = -\frac{1}{4} \\ 2B = \frac{1}{2} \Rightarrow B = \frac{1}{4} \end{cases}$$

$$\boxed{y_p = -\frac{1}{4}e^{-x} + \frac{1}{4}e^x} = \frac{1}{2} \left(\frac{e^x - e^{-x}}{2} \right) \\ \boxed{= \frac{1}{2} \sinh(x)}$$

see notes for option 2

Ex: $y^{(5)} + 2y^{(4)} - y = 3$, only find y_p

$f(x) = 3$, so we trial $y_p = A$; as long as there is no duplication with y_c .

What is y_c ?

Start with ch. eq. $r^5 + 2r^4 - 1 = 0$.

This seems hard to fully factor, but we only want to check that y_c doesn't have a piece/basis function that is $c_1 \cdot 1$ ($c_1 e^{0x}$);

so, is $r=0$ a root of char eq? *

No, b/c $0 + 0 - 1 \neq 0$, so $r \neq 0$ is not a root.

$$y_c = c_1 \overset{e^{r_1 x}}{y_1} + c_2 \overset{e^{r_2 x}}{y_2} + c_3 y_3 + c_4 y_4 + c_5 y_5$$

wanted to check if any $y_1, \dots, y_5$ could be part of an " $r=0$ piece", meaning one of them would be $= 1$ (const funct) $= e^{0x}$

so original trial ~~is~~ $y_p = A \cdot 1$ is fine.

$$\text{plug } y_p \text{ into } y_p^{(5)} + 2y_p^{(4)} - y_p = 3$$

$$\rightarrow 0 + 2 \cdot 0 - A = 3$$

$$\Rightarrow A = -3$$

so $y_p = -3$ is a particular solution

Ex: In solving the nonhomog. ODE

$$y^{(5)} - 6y^{(4)} + 9y^{(3)} = x + 2e^{3x},$$

what is the appropriate form of y_p ?

Step 1 $y_c = ?$ char. eq $r^5 - 6r^4 + 9r^3 = 0$

$$r^3(r^2 - 6r + 9) = 0$$

$$r^3(r-3)^2 = 0$$

roots are $r=0$ (mult. 3)

$r=3$ (mult. 2)

gen solution for assoc. homog. eq (y_c)

$$y_c = \underbrace{c_1 \underline{1}}_{y_1} + \underbrace{c_2 \underline{x}}_{y_2} + \underbrace{c_3 \underline{x^2}}_{y_3} + \underbrace{c_4 \underline{e^{3x}}}_{y_4} + \underbrace{c_5 \underline{x e^{3x}}}_{y_5}$$

Step 2 trial y_p : $f(x) = x + 2e^{3x}$

initially: try $y_p = \underline{A+Bx} + \underline{C e^{3x}}$

check for duplication w/ y_c .

• $A+Bx$ has duplication with y_c (spec. y_1, y_2)

$\Rightarrow$ change this to $Ax + Bx^2$

check again. Still duplication w/ y_c (y_2, y_3)

$\Rightarrow$ change to $Ax^2 + Bx^3$... still dup.

$\Rightarrow$ change to $Ax^3 + Bx^4$, now no duplication ✓

• $C e^{3x}$ piece has dupl. with y_c (spec. y_4)

$\Rightarrow$ change to $C x e^{3x}$... still duplication!

$\Rightarrow$ change to $C x^2 e^{3x}$... no duplication.

In total, trial solution $y_p = Ax^3 + Bx^4 + Cx^2 e^{3x}$.